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'Tripod-like' lung-targeting (LuT) lipids for highly efficient and selective LNPs for gene delivery and editing

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Developing lung-targeting delivery systems is essential for treating pulmonary conditions such as genetic respiratory diseases, infections, fibrosis and cancer. We synthesized and evaluated 444 lung-targeting lipids (LuT lipids) that form lipid nanoparticles (LNPs) to efficiently deliver messenger RNA and CRISPR–Cas9 genome editors to lungs with minimal side effects. Empirical analyses revealed structure–activity relationships, with top-performing LuT lipids possessing a unique 'tripod-like' structure consisting of a quaternary amine head, three long alkyl chains as legs and a short chain as a handle. LuT lipids improved endosomal escape, cargo release and endogenous targeting via adsorption of plasma proteins. Lead 1A7B13 LNPs showed a 25.5-fold improvement in mRNA delivery and a 9.2-fold increase in CRISPR–Cas9 gene-editing efficiency compared to benchmark DOTAP SORT LNPs, achieving over 90% selectivity to the lungs. 1A7B13 LNPs effectively delivered IL-10 mRNA in a therapeutic model of acute lung injury. This study reveals the relationship between lipid structure and lung-targeting activity, enriching the toolkit for lung-specific carriers.

Recent success of messenger RNA vaccines[1–3] and genome editing therapies[4–7] relied on the development and engineering of lipid nanoparticle (LNP) based delivery systems that effectively protect and deliver nucleic acids to cells[8–13]. The continued advance of RNA therapeutics requires innovation in the chemical design of improved delivery carriers for extrahepatic tissues with increased tolerability and selectivity. In particular, an array of pulmonary ailments including genetic respiratory diseases, cancer, infections and fibrosis could be treated using genetic medicines. While progress has been made in local[14–16] and systemic delivery[17–19] to the lungs, de novo design and establishment of clear structure–activity relationships (SARs) remain elusive. Moreover, the permanently cationic lipids used for selective organ targeting (SORT)[20] for lung targeting are currently limited to about 12 commercially available compounds[12,18,20,21], greatly hindering the progress towards improving potency, selectivity and safety. These factors present an opportunity for biomedical engineering.

In this Article, we present a comprehensive exploration of the design, chemical synthesis and biomedical evaluation of two libraries, comprising a total of 444 quaternary amino lipids by one-step

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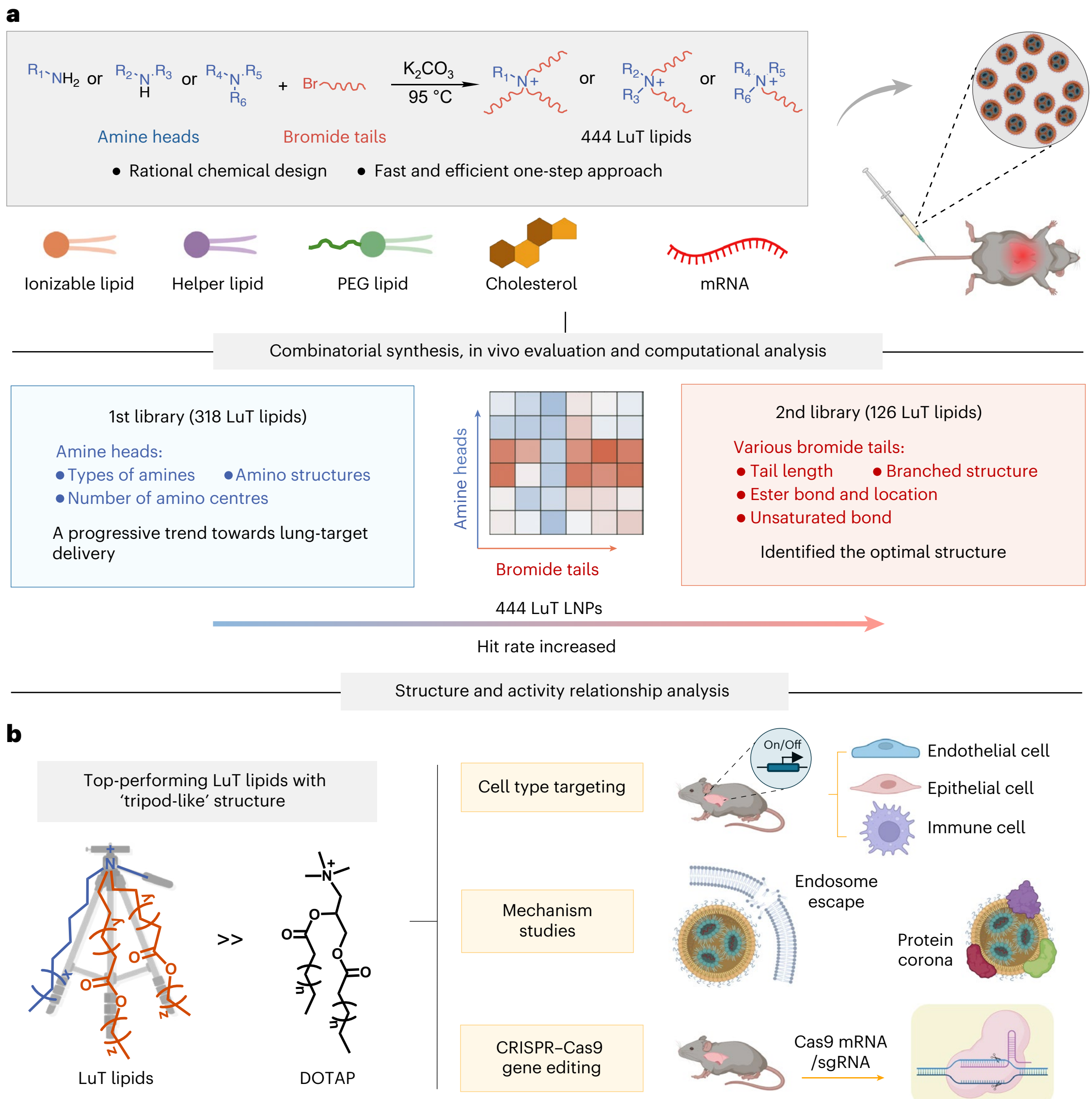

Fig. 1 | Schematic illustration of the design, synthesis and multi-round evaluation of LuT lipids tailored for effective mRNA delivery and gene-editing targeting the lungs. a, A total of 444 LuT lipids were chemically synthesized by one-step combinatorial conjugation of diverse amine heads and alkyl/alkenyl bromide tails, formulated into LuT LNPs and intravenously administrated into mice for in vivo evaluation. SARs governing the function of LuT LNPs were carefully analysed. **b**, After thorough in vivo evaluation, the top-performing LuT lipids with a tripod-like structure were identified and showed exceptional mRNA delivery efficiency, which inspired the in-depth investigation into their performance in specific cell type targeting, mechanisms and gene editing. Illustrations in **a** and **b** created in BioRender; Siegwart, D. https://BioRender.com/4609qdm (2025).

combinatorial conjugation of alkyl and alkenyl bromides with primary, secondary and tertiary amines (Fig. 1a and Supplementary Fig. 1). The resulting lung-targeting lipids (LuT lipids), composed of diverse amine heads and tails, elucidated intricate SARs governing the function of quaternary ammonium lipids. Stemming from careful in vivo evaluation, top-performing LuT molecules featured a distinctive 'tripod-like' structure, with one quaternary amine head, three long alkyl chains as legs and one short chain (typically a methyl group) as the handle (Fig. 1b). Steady improvements in hit rates across libraries yielded LuT LNPs with improved properties. In the first library, 124 out of 318 candidates exhibited promising performance (hit rate, 39%), followed by 72 out of 126 candidates in the second library (hit rate, 67%). This optimization process identified top LuT LNPs showing 25.5-fold increase in mRNA delivery efficacy compared to benchmark 1,2-dioleoyl-3-trimethylammonium-propane (DOTAP) SORT LNPs at the same administrated dose, as well as a lung delivery selectivity of more than 90%. Dimensionality reduction algorithm analysis, together with mechanism studies, helped to support potential structure–activity correlations between lipid chemistry and lung-targeting performance. The top-performing LNPs achieved highly enhanced transfection in all major cell types across the lungs, exhibiting distinct cell type enrichment for each LuT LNP. Lead LNPs based on the LuT lipid 1A7B13 enabled the most efficient transfection in epithelial cells, and 1A8B25 LNPs tended to transfect more immune cells. Lead 1A7B13 LNPs mediated CRISPR–Cas9 (clustered regularly interspaced short palindromic repeats and CRISPR-associated protein 9) genome editing 9.2-fold better than previous benchmarks and showed a highly promising therapeutic approach for treating acute lung injury (ALI) by delivering interleukin-10 (IL-10) mRNA. This study greatly expands the toolbox of lung-specific carriers and is anticipated to aid the development of safer protein replacement and gene correction therapeutics in the lungs.

Results

Design and synthesis of the first LuT lipid library (318 lipids)

To expand and diversify lung-targeting compounds, an initial library of 318 quaternary amino lipids (Fig. 2a) was synthesized using combinatorial reaction of 46 amines (1A1–2A13) and 7 alkyl bromides (B1–B7) with diverse substructural features: (1) each lipid contained at least one hydrophobic alkyl tail with chain lengths ranging from 8 to 20 carbons (Supplementary Fig. 2); (2) polar head groups exhibited one (1A) or two (2A) cationic amines, providing lipid products with one or two quaternary amine centres; (3) head groups covered a wide array of primary, secondary and tertiary amines with R groups varying from linear or cyclic, to hydroxyl or phenyl structures; (4) for 2A amine heads, the linkages between two amines included short (–CH_2CH_2–) or long (–$CH_2(CH_2)_6CH_2$–) spacers, polar (ether) or nonpolar (aliphatic) characteristics, rigid (piperazine) or flexible (methylene chain) components. This efficient chemistry approach, coupled to subsequent precipitation for effective product isolation, enabled rapid and modular synthesis of a broad and diverse collection of purified lipids.

In vivo evaluation of the first LuT lipid library revealed a progressive trend towards increased lung-targeted delivery

We formulated each LuT lipid with the ionizable lipid 4A3-SC7 (ref. 22), 1,2-dioleoyl-*sn*-glycero-3-phosphoethanolamine (DOPE), cholesterol and 1,2-dimyristoyl-*rac*-glycero-3-methoxypolyethylene glycol-2000 (DMG-PEG2000) at a molar ratio of 11.90/11.90/23.82/2.38/50 by vortex mixing[23]. This pre-optimized ratio was selected for ideal comparison to the benchmark lung SORT formulation containing 50 mol% of DOTAP[20,23].

Quantification of luciferase (Luc) activity following LNP-mediated mRNA delivery in vivo clearly reflected the impact of the chemical structures of LuT lipids on efficiency and selectivity. Among the initial 318 LuT LNPs, 89 showed higher luciferase production in lungs compared to the established benchmark lung SORT LNPs with DOTAP (Fig. 2b). As the alkyl tail lengths increased (from B1 to B7), the protein expression distribution gradually shifted from the liver to the lungs (Fig. 2c–e and Supplementary Figs. 3 and 4), with representative bioluminescent images shown in Fig. 2d. Specifically, LNPs with B1–B3 tailed lipids showed a propensity for liver tropism instead of a lung-predominant delivery profile, while B5–B7 tailed lipids showed a tendency to the lungs. While we observed a moderate hit rate (defined as the percentage of LNPs with lung-targeting selectivity surpassing 60% in the respective group) of 20% in the B4 group, there was an elevation in the B5, B6 and B7 groups with hit rates reaching 76%, 69% and 91%, respectively (Fig. 2c,e and Supplementary Fig. 5). These results highlighted the critical role of lipid tail lengths in directing the lung-targeting delivery capabilities. A chain of ~16 carbons and longer was essential for heightened lung selectivity.

SARs between amino headgroups and in vivo lung-targeting efficiency emerged (Fig. 2b,c and Supplementary Fig. 6). The inclusion of hydroxyl and aromatic groups to the lipid head did not enhance lung-specific delivery (Supplementary Fig. 6a,b); for lipid heads with carbon rings, no statistically significant differences were observed among the 5-, 6- and 7-membered rings (1A25–1A28); introducing polarity (oxygen) into the rings did not improve the lung-targeting performance. For the two-amine-head lipids, further SAR analysis revealed that variations in the number of methylene spacers between the two amines modulated lung-targeting efficiency. No significant difference was observed between lipids containing two versus three methylene groups (2A4 versus 2A5), whereas extending the spacer to eight methylene groups (2A13) significantly enhanced delivery efficiency compared to two methylenes (2A1) (Supplementary Fig. 6e,f). In addition, the presence of aromatic rings in the amine headgroup (2A11 versus 2A10) and changes in the number of hydrophobic tails (6, 4, 3 or 2) did not significantly affect lung-targeting efficiency (Supplementary Fig. 6g,h). Despite increased charges that could improve cellular uptake, the presence of multiple positively charged centres in the head groups did not improve delivery. While lipids with primary and secondary amine headgroups exhibited slightly better lung delivery outcomes compared to more complex amine types, the impact of amine head structures appeared to be less pronounced than that of bromide tails (Fig. 2f) in shifting organ tropism.

Rational design and evaluation of the second LuT lipid library (126 lipids) identified optimal lipids for effective lung-targeted delivery

Recognizing the pivotal role of the tails over the amine heads, our emphasis in building the second library was to rationally incorporate diverse variations in the tail structures (Fig. 3a). Specifically, with a careful selection of seven amine heads from the first library (Supplementary Fig. 7), we first increased the hydrocarbon tail lengths (for example, B8) and included branched structures (for example, B13–B18). Second, to enhance the biocompatibility, we introduced degradable ester linkages in the tails by the esterification of bromoacetic acids and alkyl or alkenyl alcohols (Supplementary Fig. 8) to allow enzymatic hydrolysis, which released a moderate zwitterionic lipid, along with an alcohol metabolite in tissues for easier clearance (Supplementary Fig. 9)[24]. Meanwhile, we modulated the position of the ester bond, which was reported to provide the ideal balance of in vivo lipid clearance and protein expression[25]. Finally, we incorporated unsaturated bonds with varying numbers (one (B19, B23, B24, B25) or multiple (B20, B21 and B22)) and positions (at the end or in the middle of the tail).

SAR analysis indicated the introduction of branched structures and increased unsaturation in the tails improved the lipid performance (Fig. 3b–f). In particular, the ester bond position proved crucial. As the ester bond moved closer to the nitrogen, the lung delivery efficiency decreased, evidenced by lipids with B11, B16 and B20 tails (Fig. 3b–d,g and Supplementary Fig. 10) where the quaternary amine nitrogen and ester were less than three carbons apart. This ester placement within the hydrophobic domain may sterically hinder the molecular geometry required for lipid nanoparticle self-assembly.

Exploring these two libraries, we observed that most of the obtained 444 LuT LNPs exhibited suitable hydrodynamic diameters of 100–150 nm, low polydispersity index (PDI) less than 0.2 and high encapsulation efficiency (Fig. 3h and Supplementary Figs. 11–13). In vitro cytotoxicity studies showed that the majority of LuT LNPs exhibited minimal toxicity (Supplementary Fig. 14). Attributed to a more stringent design, nearly half of the LNPs in the second library exhibited improved selectivity and luciferase production, with 17 outperforming LNPs achieving over a 10-fold higher mRNA-mediated protein expression in the lungs compared to control DOTAP SORT LNPs (Fig. 3b,c,i and Supplementary Figs. 15 and 16). Together with the 8 superior lipids selected from the first library, we identified 25 LuT lipids in total for further investigation (Fig. 3i). Most lipids possessed 1A7 and 1A8 amine heads and exhibited high selectivity with >90% of the total signal expressed in the pulmonary region (Fig. 2b and 3b,i). We noticed that most of these top-performing LuT lipids featured in a distinctive tripod-like structure, characterized by one quaternary amine head, three long alkyl chains as the 'stands' and one short chain (typically a methyl group) as the 'handle' (Fig. 3j,k). Tripod lipids include those that contain head groups 1A1, 1A2, 1A3, 1A6, 1A7 or 1A8 in combination with tails B4–B25. *t*-Distributed stochastic neighbour embedding dimensionality reduction clustering analysis based on 12 structural descriptors confirmed a group of high-performing lipids sharing the 'tripod' structural motif, consistent with in vivo findings (Supplementary Fig. 17). clogP (calculated logP) exhibited a statistically positive significant correlation with lung-targeting selectivity (Fig. 3l).

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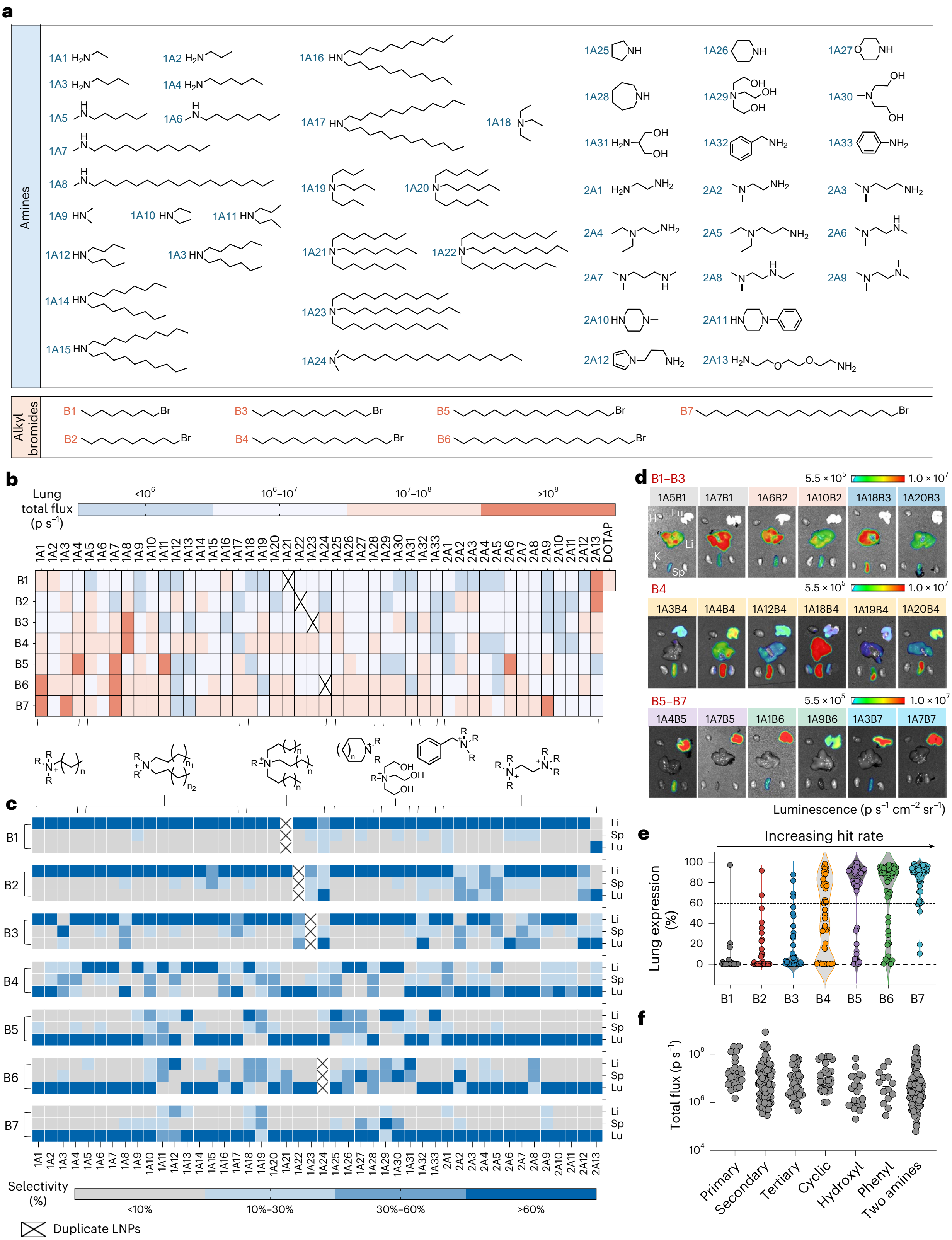

Fig. 2 | The first library of LuT LNPs exhibited a progressive trend towards increased lung-targeting delivery. a, Chemical structures of the amines and alkyl bromides used to build the LuT lipids in the first library. **b**, Heat map of the total luciferase expression in the lungs following i.v. administration of LuT LNPs containing luciferase mRNA (0.1 mg kg^{-1} Luc mRNA, 6 h) into C57BL/6 mice. p s^{-1}, photons per second. **c**, Corresponding relative luciferase activity selectivity in major organs including liver, lungs and spleen. Li, liver; Sp, spleen; Lu, lung. **d**, Representative bioluminescence images of the major organs of mice. H, heart; K, kidney. **e**, The expression percentage in the lungs increased as the tail length increased from B1 to B7. Lung-targeting selectivities >60% were defined as hits for hit rate calculation. **f**, Total luciferase activity in the lungs induced by LuT LNPs with different types of amine heads.

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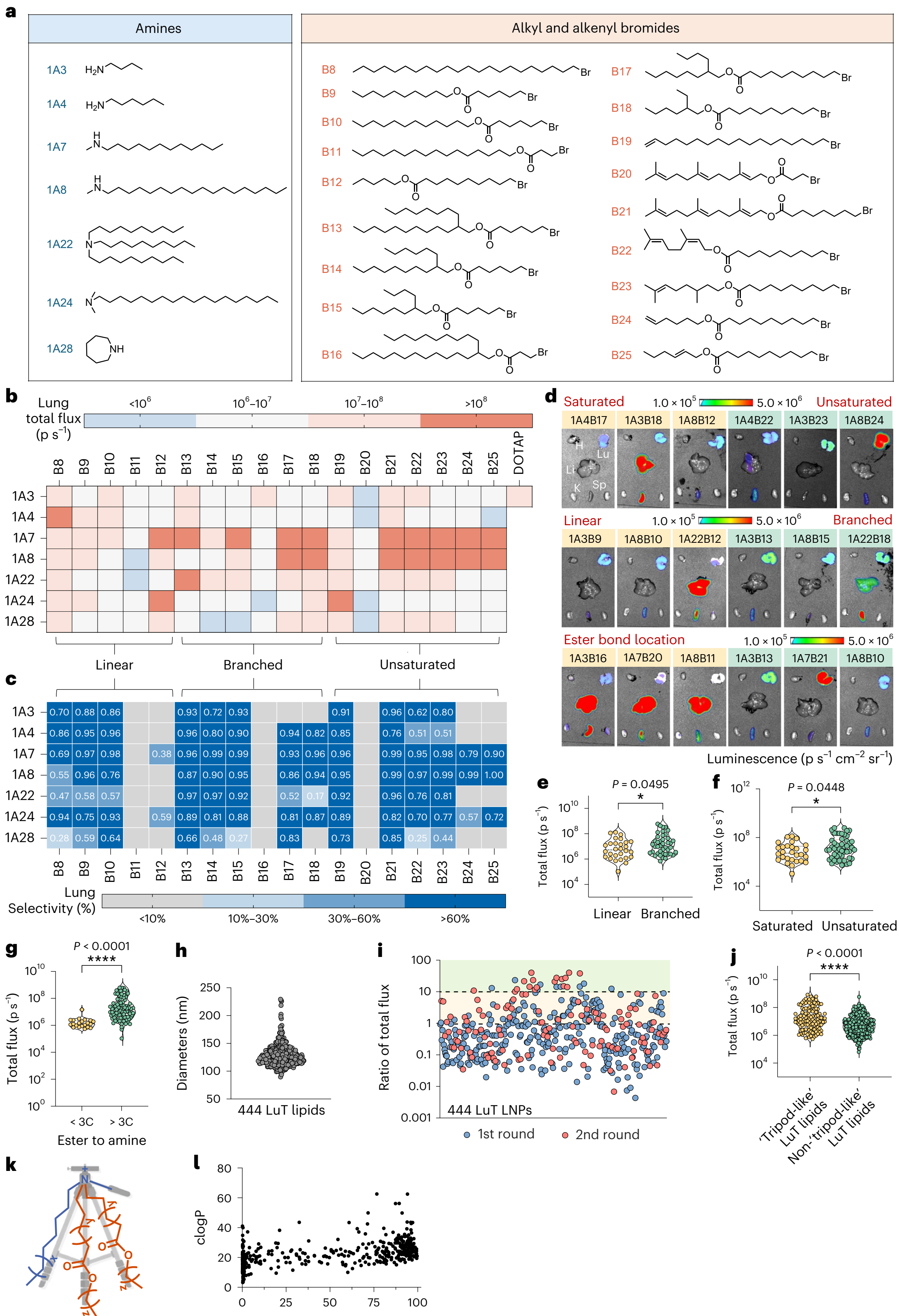

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Fig. 3 | Rational design and evaluation of the second library of LuT LNPs identified the optimal tripod-like structures for exceptional lung-targeted delivery. **a**, Chemical structures of the amines and alkyl/alkenyl bromides used to build the LuT molecules in the second library. **b**,**c**, Heat maps of the total luciferase expression (**b**) and relative luciferase expression selectivity (**c**) in the lungs following i.v. administration of LuT LNPs containing luciferase mRNA (0.1 mg kg^{-1} Luc mRNA, 6 h) into C57BL/6 mice. **d**, Representative bioluminescence images of the major organs of mice. **e**–**g**, Total luciferase activity in the lungs induced by LuT LNPs with linear or branched tails (**e**), saturated or unsaturated tails (**f**) and different positions of the ester bond in the tails (**g**). A two-tailed unpaired *t*-test was used to determine the statistical significance of the comparisons shown in **e**–**g**. $^{*}P < 0.05$, $^{****}P < 0.0001$. **h**, Hydrodynamic diameters of 444 LuT LNPs obtained by dynamic light scattering, most of which exhibited a suitable size of 100–150 nm. **i**, The ratios of the lung signals induced by 318 LuT LNPs in the first library and 126 LuT LNPs in the second library to DOTAP LNPs. The 25 top-performing LuT lipids showing >10-fold improvement in luciferase expression over DOTAP were selected. **j**, Total luciferase activity in the lungs induced by LuT LNPs containing LuT lipids with a tripod-like structure (one quaternary amine head, three long alkyl chains and one short chain) or non-tripod-like structures. Statistical significance was determined using a two-tailed unpaired *t*-test. $^{****}P < 0.0001$. **k**, Top-performing LuT lipids showed a distinctive tripod-like structure, characterized by three long alkyl chains as the stands and one short chain (typically a methyl group) as the handle. **l**, The correlation between clogP and the lung-targeting selectivity shows a statistically positive correlation ($r = 0.53$, $P < 0.0001$). Statistical significance was assessed using a two-sided Pearson correlation test.

Top LuT LNPs exhibited superior lung-specific delivery potency and outstanding Cre-mediated recombination in different lung cell types

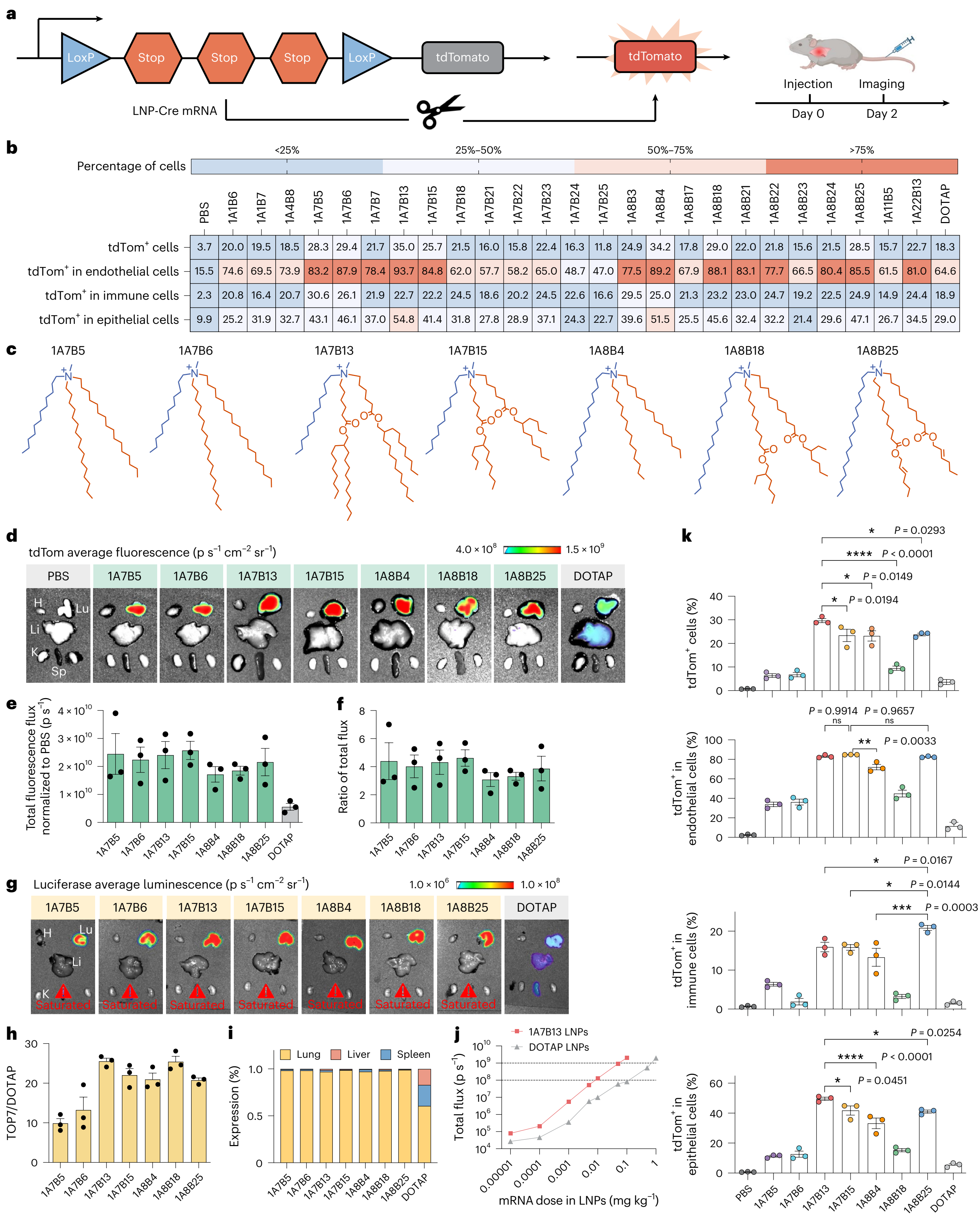
To quantify the ability of LuT LNPs to mediate lung-specific Cre recombinase (Cre)-mediated recombination, we used a genetically engineered Cre-LoxP (locus of X-over P1) mouse model (Ai14) containing LoxP-flanked 'stop' sequences positioned upstream of the tdTomato gene. Once Cre mRNA is successfully delivered and expressed, the stop cassettes are deleted, and tdTomato fluorescence is turned on and detected in cells (Fig. 4a). Following a single dose of 0.3 mg kg^{-1} Cre mRNA[5,26], 7 top hits were identified from the 25 LuT LNPs with favourable features for effective lung targeting (Supplementary Fig. 18), based on total tdTomato^{+} (tdTom^{+}) cells (Fig. 4b and Supplementary Fig. 19)[10,21]. The 7 LuT lipids were all composed of 1A7 or 1A8 heads, and two additional hydrophobic domains (Fig. 4c and Supplementary Figs. 20–26). The top-performing LNPs all induced superior lung-selective Cre recombination compared to the benchmark DOTAP control (Fig. 4d–f), with a 3.2- to 5.9-fold increase in total lung fluorescent signal and 75–90% of the signal distributed in the lungs, whereas the value for DOTAP LNPs at the same dose was only 36% (Supplementary Fig. 27). Similarly, when delivering Luc mRNA (0.1 mg kg^{-1}), the top 7 LuT LNPs exhibited up to ~25.5-fold higher signal (Fig. 4g,h and Supplementary Fig. 28), with >95% of lung-delivery selectivity and specificity, surpassing DOTAP LNPs (Fig. 4i). Using Cy5-labelled mRNA to track biodistribution, 1A7B13 LNPs were more highly enriched in the lungs compared to benchmark DOTAP LNPs (Supplementary Fig. 29). We further conducted an in vivo dose–response experiment to provide a direct comparison of the efficacy of 1A7B13 LNPs versus DOTAP LNPs, from 0.00001 to 1.0 mg kg^{-1} Luc mRNA (Fig. 4j). A dose-dependent increase in luciferase activity was observed for both formulations. At same doses, the delivery efficiency of 1A7B13 LNPs was 10 to 25 times greater than that of DOTAP LNPs (Fig. 4j). At equivalent levels of luciferase expression, DOTAP LNPs required approximately tenfold higher mRNA doses compared to 1A7B13 LNPs (Supplementary Fig. 30a,b). In addition, it was observed that the luminescent signal was saturated at 0.1 mg kg^{-1} mRNA dose for 1A7B13 LNPs, whereas DOTAP LNPs required 1.0 mg kg^{-1} mRNA dose to achieve comparable levels of activity (Supplementary Fig. 30c,d).

To provide a guide in using different LuT lipids to address lung diseases arising from endothelial, epithelial, immune or other cell types[27–30], we further examined the selectivity of specific cell types in the lungs induced by each individual LNP. In general, the top 7 LuT LNPs transfected 30% of the total lung cells on average overall, which was a notable improvement compared to 18.3% by DOTAP LNPs (Fig. 4b) at Cre mRNA dose of 0.3 mg kg^{-1}. LuT LNPs transfected approximately 80–90% of all endothelial cells, 40–50% of all epithelial cells and 25% of immune cells across the lungs, outperforming the benchmark (Fig. 4b). The average ratio of tdTom^{+} epithelial cells to tdTom^{+} endothelial cells for the top 7 LuT LNPs was 0.59, versus 0.45 for DOTAP LNPs (Supplementary Fig. 31), suggesting that LuT lipids could mediate mRNA delivery to deeper lung tissues. More specifically, we observed high tdTom expression in endothelial cells (93.7% tdTom^{+}) and epithelial cells (54.8% tdTom^{+}) induced by the lead 1A7B13 LNPs. To better evaluate whether the top LuT LNPs exhibit cell type-specific delivery advantages, we reduced the Cre mRNA dose to 0.05 mg kg^{-1}. As shown in Fig. 4k, 1A7B13 LNPs enabled the most efficient transfection in epithelial cells (49.57% tdTom^{+}), while 1A8B25 LNPs tended to transfect more immune cells (20.83% tdTom^{+}). This allowed us to identify appropriate LuT molecules that are preferentially delivered to specific lung cell types. Moreover, all top 7 LuT LNPs elicited no detectable inflammatory response following administration (Supplementary Fig. 32).

LuT molecules facilitated endosomal escape and promoted interactions with specific plasma proteins for optimal lung-targeting delivery

Similar to our previous results[31], there were no obvious correlations between the LNP physicochemical properties (size, PDI, zeta potential, encapsulation efficiency) and their improved delivery performance

Fig. 4 | Top LuT LNPs exhibited superior organ- and cell-specific delivery potency. **a**, Schematic showing the delivery of Cre mRNA activates tdTomato expression in tdTomato transgenic mice via Cre-mediated genetic deletion of the stop cassette. **b**, Fluorescence-activated cell sorting (FACS) quantification of the tdTom^{+} cell percentages in endothelial (CD31^{+}), epithelial (EpCAM^{+}) and immune cells (CD45^{+}) across the lungs of tdTomato mice following a single dose of 25 LuT LNPs containing Cre mRNA (0.3 mg kg^{-1} Cre mRNA, 48 h). **c**, Top 7 LuT lipids were identified. **d**, Ex vivo fluorescence images of the major organs from tdTomato mice after i.v. injection of top 7 LuT LNPs containing Cre mRNA (0.3 mg kg^{-1} Cre mRNA, 48 h). **e**, Top 7 LuT LNPs all induced a superior efficient and selective Cre mRNA delivery to lungs, compared to benchmark DOTAP LNPs. **f**, The ratio of tdTomato fluorescence signal induced by top 7 LuT LNPs to that induced by DOTAP LNPs. Data are shown as mean ± s.e.m. ($n = 3$ biologically independent animals). **g**, Ex vivo luminescence images of the major organs from C57BL/6 mice after i.v. injection of top LuT LNPs containing Luc mRNA (0.1 mg kg^{-1} Luc mRNA, 6 h). **h**,**i**, Top 7 LuT LNPs showed up to 25.5-fold increase in mRNA delivery efficacy (**h**) and improved lung-delivery selectivity and specificity (**i**) compared to benchmark DOTAP LNPs at the same administration dose. **j**, Dose–response curves depicting total luminescence in the lungs following i.v. administration of Luc mRNA-loaded 1A7B13 LNPs or DOTAP LNPs with mRNA doses ranging from 0.00001 mg kg^{-1} to 1.0 mg kg^{-1} in C57BL/6 mice. Data are shown as mean ± s.e.m. ($n = 3$ biologically independent animals). **k**, FACS quantification of the percentages of tdTom^{+} cells in different cell populations within the lungs of tdTomato mice. Ai14 mice were intravenously administered with the top 7 LuT LNPs containing a low dose of Cre mRNA (0.05 mg kg^{-1}, 48 h). Total tdTom^{+} cells, tdTom^{+} endothelial cells, tdTom^{+} immune cells, tdTom^{+} epithelial cells in PBS, top 7 LuT LNPs and DOTAP-treated groups were quantified. Data are shown as mean ± s.e.m. ($n = 3$ biologically independent animals, one-way ANOVA followed by Dunnett's post hoc test was used for multiple comparisons). $^{*}P < 0.05$; $^{**}P < 0.01$; $^{***}P < 0.001$; $^{****}P < 0.0001$; ns, not significant, $P > 0.05$. Illustrations in **a** (right) created in BioRender; Siegwart, D. https://BioRender.com/4609qdm (2025).


a
LoxP
Stop
Stop
Stop
LoxP
tdTomato
tdTomato
LNP-Cre mRNA
Injection
Imaging
Day 0
Day 2
b
Percentage of cells
<25%
25%–50%
50%–75%
>75%

	PBS	1A1B6	1A1B7	1A4B8	1A7B5	1A7B6	1A7B7	1A7B13	1A7B15	1A7B18	1A7B21	1A7B22	1A7B23	1A7B24	1A7B25	1A8B3	1A8B4	1A8B17	1A8B18	1A8B21	1A8B22	1A8B23	1A8B24	1A8B25	1A11B5	1A22B13	DOTAP
tdTom+ cells	3.7	20.0	19.5	18.5	28.3	29.4	21.7	35.0	25.7	21.5	16.0	15.8	22.4	16.3	11.8	24.9	34.2	17.8	29.0	22.0	21.8	15.6	21.5	28.5	15.7	22.7	18.3
tdTom+ in endothelial cells	15.5	74.6	69.5	73.9	83.2	87.9	78.4	93.7	84.8	62.0	57.7	58.2	65.0	48.7	47.0	77.5	89.2	67.9	88.1	83.1	77.7	66.5	80.4	85.5	61.5	81.0	64.6
tdTom+ in immune cells	2.3	20.8	16.4	20.7	30.6	26.1	21.9	22.7	22.2	24.5	18.6	20.2	24.5	22.6	16.6	29.5	25.0	21.3	23.2	23.0	24.7	19.2	22.5	24.9	14.9	24.4	18.9
tdTom+ in epithelial cells	9.9	25.2	31.9	32.7	43.1	46.1	37.0	54.8	41.4	31.8	27.8	28.9	37.1	24.3	22.7	39.6	51.5	25.5	45.6	32.4	32.2	21.4	29.6	47.1	26.7	34.5	29.0

c
1A7B5
1A7B6
1A7B13
1A7B15
1A8B4
1A8B18
1A8B25
d
tdTom average fluorescence (p s−1 cm−2 sr−1)
4.0 × 10^8
1.5 × 10^9
PBS
1A7B5
1A7B6
1A7B13
1A7B15
1A8B4
1A8B18
1A8B25
DOTAP
H
Lu
Li
K
Sp
e
Total fluorescence flux normalized to PBS (p s−1)
f
Ratio of total flux
g
Luciferase average luminescence (p s−1 cm−2 sr−1)
1.0 × 10^6
1.0 × 10^8
Saturated
h
TOP7/DOTAP
i
Lung
Liver
Spleen
Expression (%)
j
1A7B13 LNPs
DOTAP LNPs
Total flux (p s−1)
mRNA dose in LNPs (mg kg−1)
k
tdTom+ cells (%)
tdTom+ in endothelial cells (%)
tdTom+ in immune cells (%)
tdTom+ in epithelial cells (%)
P = 0.0293
P < 0.0001
P = 0.0149
P = 0.0194
P = 0.9914
P = 0.9657
ns
P = 0.0033
P = 0.0167
P = 0.0144
P = 0.0003
P = 0.0254
P < 0.0001
P = 0.0451

(Supplementary Fig. 33), which prompted us to delve deep into more profound underlying mechanisms.

The effectiveness of LNP-based delivery systems hinges crucially on efficient endosomal escape[32–35]. Due to the unique tripod-like chemical structure of LuT lipids and similarity to iPhos—a phospholipid known to enhance endosomal escape through a shape-change mechanism—we hypothesized that LuT lipids could facilitate the endosomal escape and subsequent cargo release[36,37]. Formation of a cone-shaped structure may promote membrane hexagonal transformation, which is advantageous over DOTAP-like molecules with only two alkyl tails constructing a cylindrical shape (Fig. 5a)[32,34,35]. Meanwhile, instead of the close distance between ester and nitrogen in DOTAP-like molecules, LuT molecules feature esters farther away from the quaternary centre, providing more flexibility. We first conducted a fluorescence resonance energy transfer (FRET) assay to evaluate the membrane fusion and LNP dissociation process. Endosomal mimicking liposomes were co-labelled with 1,2-dioleoyl-*sn*-glycero-3-phosphoethanolamine-*N*-(7-nitro-2-1,3-benzoxadiazol-4-yl) (NBD-PE, donor fluorophore) and 1,2-dioleoyl-*sn*-glycero-3-phosphoethanolamine-*N*-(lissamine rhodamine B sulfonyl) (Rho-PE, acceptor fluorophore), and the close contact between FRET pairs led to attenuated NBD signal. Following membrane fusion, the increase in the donor–acceptor distance resulted in the recovery of NBD signal[38,39]. We noticed that the LNP-mRNA complexes promptly fused with endosomal mimicking liposomes upon initial mixing, and the percentage of fused membranes gradually increased over time (Fig. 5b). The top 7 LuT LNPs all exhibited augmentation in the membrane fusion and endosomal rupture compared to DOTAP LNPs, indicating a robust trend to disrupt endosomal membranes. In addition, LuT LNPs were easier to disassemble, facilitating the release of mRNA upon mixing with endosomal mimicking liposomes (Fig. 5c).

Extracellularly, intravenously administered LNPs bind plasma proteins, creating an interfacial layer called the 'protein corona'[21,37,40]. While the variability of biomolecular corona composition was previously recognized for its impact on LNPs' extrahepatic targeting tropism[41], further investigation is required to understand the precise relationship between specific proteins and lung-targeting outcomes. We followed an established protocol to identify plasma proteins absorbed to LNPs[21]. Sodium dodecyl sulfate–polyacrylamide gel electrophoresis (SDS–PAGE) (Fig. 5d) revealed an increased enrichment of bands between 50 and 75 kDa with the top 7 LuT LNPs compared to DOTAP LNPs, along with a modest decease in bands around 37 to 50 kDa. Liquid chromatography–tandem mass spectrometry identified and quantified corona proteins absorbed to LuT LNPs and DOTAP LNPs, generally showing negative charge at physiological pH (Supplementary Fig. 34). Vitronectin, albumin, fibrinogen, serum paraoxonase/arylesterase 1 and clusterin were consistently present in the protein coronas of evaluated LNPs (Fig. 5e–g). Hierarchical clustering analysis of the protein corona fingerprints showed that the protein coronas of LuT LNPs and DOTAP LNPs were distinguished by two distinct clusters (Supplementary Fig. 35), highlighting a difference in corona phenotypes between these two groups. LuT lipids facilitated increased absorption of vitronectin (Fig. 5e,g), a protein identified as the most crucial factor for endogenous targeting of lung cells expressing high levels of vitronectin's cognate receptor αvβ3-integrin[21,31]. The vitronectin abundance in the coronas of 1A7B5 and 1A8B18 LNPs (~40%) was 4 times higher than benchmark DOTAP LNPs (Fig. 5g). We also included an additional four LuT LNPs: 1A7B1, 1A7B2, 1A7B3 and 1A7B7 LNPs (Supplementary Fig. 36), which further indicate enhanced vitronectin binding by the LuT LNPs. These mechanistic advantages were achieved without compromising safety. No signs of thrombosis were detected in mice treated with 1A7B13 LNPs at the same or double the mRNA dose tested previously[42] (Fig. 5h,i and Supplementary Fig. 37). In addition, liver (aspartate aminotransferase (AST) and alanine aminotransferase (ALT)) and kidney (blood urea nitrogen (BUN) and creatinine (CREA)) enzyme functions and tissue histological analysis revealed no overt signs of in vivo toxicity associated with 1A7B13 LNPs (Fig. 5j and Supplementary Figs. 38 and 39). These results indicated that the unique chemical structures of LuT lipids could induce changes in the protein corona compositions, promoting binding with proteins positively correlated with lung specificity, consequently influencing the delivery outcomes.

LuT LNPs mediated increased lung-specific CRISPR–Cas9 gene editing with translational potential

Delivering gene editing tools, such as CRISPR–Cas9, to target lung cells in a safe and effective manner still remains a critical challenge[28]. We next compared the capabilities of 1A7B13 and DOTAP LNPs in co-delivering Cas9 mRNA and single guide RNA (sgRNA) for lung-specific CRISPR–Cas9 gene editing (Fig. 6a). 1A7B13 or DOTAP LNPs co-delivering Cas9 mRNA and Tom1 sgRNA (sgTom1) with a weight ratio of 2:1 were intravenously administered into Ai14 mice at a total RNA dose of 0.75 mg kg^{-1} (2 doses, 48 h apart) to target the 'Lox–stop–Lox' sequence upstream of the tdTomato gene. Cell-type-specific gene-editing efficiency in the lungs was quantified 10 days after the second administration by measuring the tdTomato fluorescence (Fig. 6b). 1A7B13 and DOTAP LNPs both induced a specific tdTom fluorescence in the lungs, while the signal enabled by 1A7B13 LNP was 5.6-fold higher (Fig. 6c), suggesting robust CRISPR–Cas9 gene editing in the lungs. The tdTom fluorescence strength was further confirmed by imaging the lung tissue sections using confocal microscopy (Fig. 6d). Flow cytometry data revealed that 1A7B13 LNPs induced tdTom expression in 26.5% of total lung cells, marking a 4.3-fold augmentation compared to 6.2% observed in DOTAP LNPs at the tested low dose. Specifically, for 1A7B13 LNP-treated mice, approximately 73% of endothelial cells, 34% of epithelial cells and 16% of immune cells within the lungs exhibited evidence of editing, versus a lower percentage of 31% endothelial, 23% epithelial, and 6% immune for DOTAP LNP-treated mice (Fig. 6e). Next we investigated the editing efficiency of LuT LNPs for an endogenous target using the tumour suppressor gene phosphatase and tensin homologue (PTEN), which is expressed in most cells[43]. Wild-type C57BL/6 mice were intravenously injected with 2 doses of 1A7B13 or DOTAP LNPs encapsulating Cas9 mRNA/sgPTEN (2:1 weight ratio; 0.75 mg kg^{-1} total RNA). Insertions

Fig. 5 | LuT molecules facilitated endosomal escape and promoted the interactions with specific plasma proteins for optimal lung-targeting delivery. a, Possible cone-shaped structure formed by a representative LuT molecule and cylindrical structure by DOTAP. **b**, Lipid fusion and membrane rupture of LuT LNPs and DOTAP LNPs determined by a FRET assay at pH 5.5. Data are shown as mean ± s.e.m. (*n* = 3 biologically independent samples). **c**, LuT LNPs and DOTAP LNPs dissociation characterized by FRET assay after mixing with endosomal mimics for 10 min at pH 5.5. Data are shown as mean ± s.e.m. (*n* = 3 biologically independent samples). **d**, SDS–PAGE of the protein coronas of LuT LNPs. Vtn, vitronectin. **e**, Protein corona fingerprints of LuT LNPs (*n* = 3 biologically independent animals). **f**, Classification of the most abundant proteins in coronas of LuT LNPs by functions (Supplementary Table 1). **g**, Compositions of the eight most abundant factors from the protein coronas of LuT LNPs compared with DOTAP LNPs. Data are shown as mean ± s.e.m. (*n* = 3 biologically independent samples). **h**, Hydrodynamic diameters of mRNA-loaded 1A7B13 LNPs obtained by dynamic light scattering. **i**, Histological analysis of the lung tissues showed that DOTAP lipoplexes induced the formation of large clots in the pulmonary vasculature and capillaries, as indicated by the arrows (scale bar, 100 μm). No observable clots were detected in mice treated with 1A7B13 LNPs. **j**, Intravenous administration of 1A7B13 LNPs encapsulating Luc mRNA was well tolerated in C57BL/6 mice (0.75 mg kg^{-1}, *n* = 3 biologically independent animals). LPS (5 mg kg^{-1}, intraperitoneal administration) was used as a positive control, and PBS (i.v.) was used as a negative control. Tissue sections of the lungs were collected 48 h after injection and stained with H&E. Scale bar, 100 μm. Illustrations in **b** (left) and **c** created in BioRender; Siegwart, D. https://BioRender.com/4609qdm (2025).

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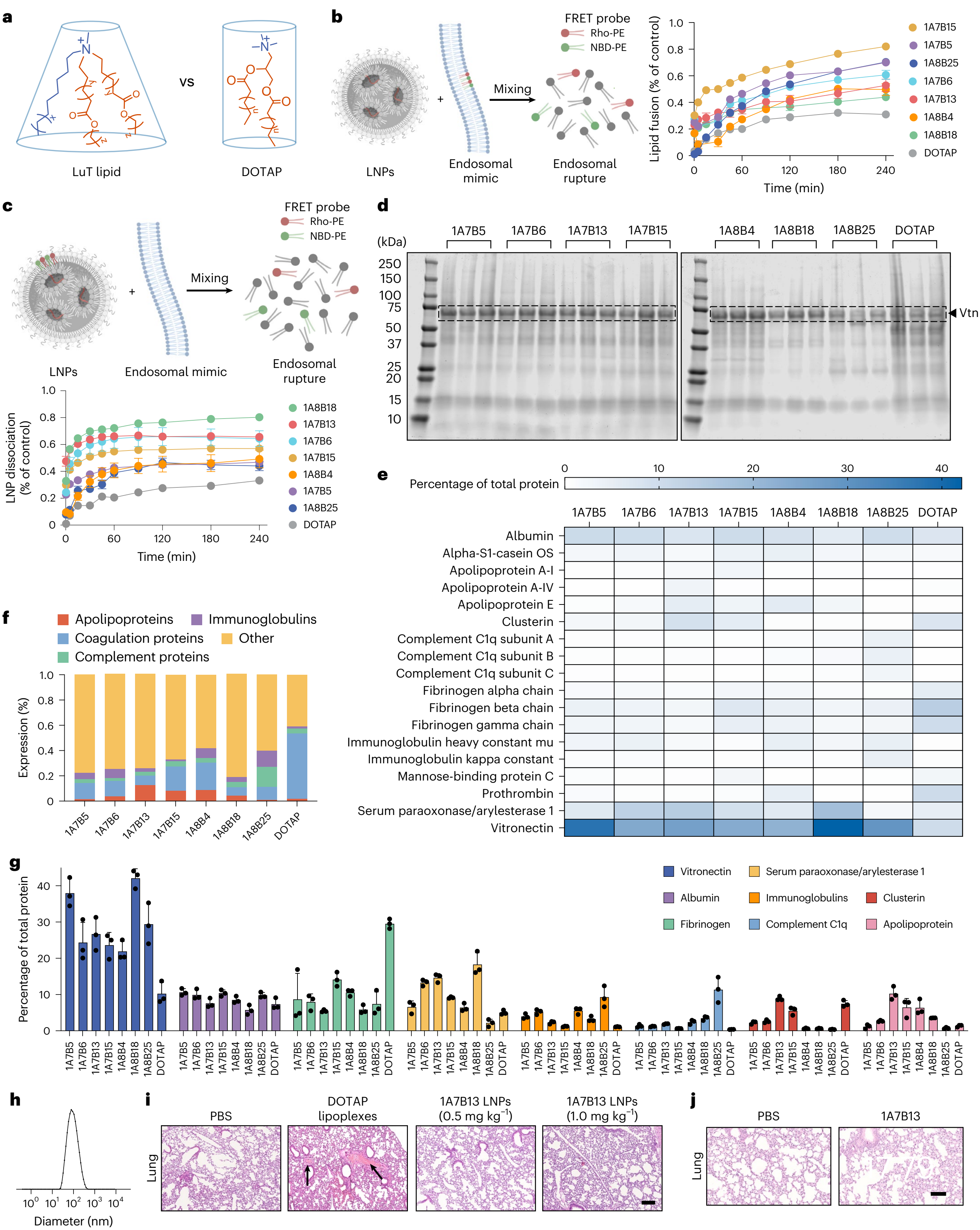

and deletions (indels) were quantified after 12 days (Fig. 6f). The T7 endonuclease 1 (T7E1) assay detected clear DNA cleavage bands in the 1A7B13 group, with only minimal cleavage bands in the DOTAP group at equivalent treatment doses (Fig. 6g). Sanger sequencing analysis of total lung genomic DNA revealed the presence of 20.23% indels at the PTEN locus in 1A7B13-treated mice, compared to only 2.20% in DOTAP-treated mice, showing a 9.2-fold improvement of the gene-editing efficiency (Fig. 6h). Consistently, immunohistochemistry (IHC) staining indicated that 67.4% of lung cells in the 1A7B13 group were PTEN negative, which was higher than 27.4% in the DOTAP group (Fig. 6i and Supplementary Fig. 40). Haematoxylin and eosin (H&E) staining also showed a clearer cytoplasm in the lung cells from the 1A7B13 group, which is a known phenotype of PTEN loss due to lipid accumulation[20,43] (Fig. 6i). Overall, these data show that LuT LNPs, exemplified by 1A7B13, largely increased CRISPR–Cas9 gene editing across diverse cell types within the lungs, showing promising potential for future applications while maintaining good tolerance.

1A7B13 LuT LNPs can deliver IL-10 mRNA to treat ALI

Given the high transfection efficiency of the lead 1A7B13 LNPs in both lung endothelial and epithelial cells, we proceeded to assess their therapeutic potential in delivering IL-10 mRNA to treat ALI. C57BL/6 mice were intravenously administered 1A7B13 LNPs or DOTAP LNPs encapsulating IL-10 mRNA (0.5 mg kg^{-1}) on days −1, 0 and 1. High-dose lipopolysaccharide (LPS, 10 µg) was given on day 0 through oropharyngeal aspiration (OPA) to induce the ALI model. Mice were killed on day 2 for analysis (Fig. 6j). Compared to the DOTAP group, treatment with 1A7B13 LNPs significantly attenuated LPS-induced acute weight loss by day 2 (Fig. 6k,l). Histopathological analysis of lung tissues revealed a marked reduction in LPS-induced interstitial congestion and inflammation in mice treated with 1A7B13 LNPs, compared to the phosphate buffered saline (PBS) and DOTAP groups (Fig. 6m). This observation was further confirmed by semi-quantitative assessment conducted by a pathologist blinded to the treatment conditions (Fig. 6n), corresponding to a decrease in overall total lung cell counts with 1A7B13 treatment (Fig. 6o). In addition, mice treated with 1A7B13 LNPs exhibited significantly elevated IL-10 mRNA expression in the lungs (Fig. 6p) and IL-10 protein levels in bronchoalveolar lavage fluid (BALF) (Fig. 6q). Meanwhile, total protein concentrations in BALF, a marker of lung permeability and lung injury[44], were markedly reduced in 1A7B13-treated mice compared to the DOTAP group (Fig. 6r). To further evaluate the severity of inflammation, mRNA levels of key airway inflammatory markers, including TNF, IL-1β and IL-6, were measured (Fig. 6s–u). Mice treated with 1A7B13 LNPs exhibited relatively lower expression levels compared to the DOTAP group, indicating a mitigated inflammatory response. This mitigation is attributed to the targeted delivery and localized expression of anti-inflammatory IL-10 in the lungs, which achieved therapeutically relevant cytokine concentrations at the site of injury. The administration of 1A7B13 LNPs greatly alleviated lung pathology and effectively reduced the severity of ALI, underscoring its potential as an effective intervention for ALI.

To further assess the versatility of 1A7B13 LNPs beyond mRNA delivery, we also evaluated their efficacy in lung-specific Tie2 (tyrosine kinase with immunoglobulin-like and epidermal growth factor-like domains 2) small interfering RNA delivery. 1A7B13 LNPs achieved higher knockdown of Tie2 than DOTAP LNPs at the same dose, with increased phenotypic changes resulting from the higher potency (Supplementary Fig. 41). These results highlight applicability of LuT LNPs siRNA applications, showing their broad potential.

Discussion

The use of lung-targeting LNPs for mRNA delivery and genome editing holds great promise to treat various pulmonary diseases. Motivated by the limited commercial availability of lung-targeting lipids and lack of in-depth studies on the SAR, we rationally designed and synthesized 444 quaternary amino lipids (LuT lipids) with diverse functionalities tailored for lung-targeting delivery. Thorough analysis of in vivo empirical data we identified the top-performing LuT lipids with a tripod-like structure with one single quaternary amine head, three long hydrophobic domains and one short methyl chain. LuT LNPs showed an enhancement of protein expression (up to a 25.5-fold improvement) and high lung selectivity (90%) compared to our previous leading DOTAP LNPs used in our recent work that achieved long-term gene editing of lung stem cells[45]. The representative 1A7B13 LNPs increased the lung-selective CRISPR–Cas9 gene editing by 9.2-fold. The unique structure of LuT lipid enabled superior lung-targeting potency and specificity. Unlike DOTAP molecules, where only two alkyl tails constituted a cylindrical shape and bulky ester groups were close to the nitrogen, LuT molecules formed a cone-shaped structure with more flexibility that promoted membrane hexagonal formation, facilitating endosomal escape and subsequent cargo release[36]. The distinctive LuT molecules also helped LNPs to bind specific proteins positively related to lung delivery, forming protein coronas enriched in factors such as vitronectin. These advantages collectively produced a synergistic impact that drove to the enhanced lung-targeting efficiency and specificity.

Fig. 6 | Representative 1A7B13 LNPs mediated highly enhanced lung-specific CRISPR–Cas9 gene editing and superior ability to treat ALI. a, 1A7B13 LNPs with tripod-like structure hold potential to achieve effective CRISPR–Cas9 gene editing. **b**, The schematic illustration shows that co-delivery of Cas9 mRNA and sgTom1 activates tdTomato expression in tdTomato transgenic mice. **c**, Ex vivo fluorescent images of the major organs from tdTomato mice on day 12 after i.v. administration of 1A7B13 LNPs or DOTAP LNPs containing Cas9 mRNA/sgTom1 (0.75 mg kg^{-1} Cas9 mRNA/sgTom1, two doses on days 0 and 2). **d**, Confocal microscopy images showed an obvious strength of $tdTom^{+}$ cell signal in the lung tissues after being administrated with LuT LNPs. Scale bar, 100 µm. **e**, FACS quantification of the $tdTom^{+}$ cell percentages in endothelial ($CD31^{+}$), epithelial ($EpCAM^{+}$) and immune cells ($CD45^{+}$) across the lungs of tdTomato mice following two doses of LNPs containing Cas9 mRNA/sgTom1. Data are shown as mean ± s.e.m. ($n = 3$ biologically independent animals; a two-tailed unpaired *t*-test was used to determine the statistical significance of the comparison). **f**, LNPs were used to co-deliver Cas9 mRNA and sgPTEN to C57BL/6 mice (0.75 mg kg^{-1} Cas9 mRNA/sgPTEN, i.v., two doses on days 0 and 2, 14 days). **g**, T7E1 assays indicated that lung-specific PTEN editing was achieved ($n = 3$ biologically independent animals). **h**, Gene-editing efficiency was calculated by tracking of indels by decomposition (TIDE) analysis. Indels at the PTEN locus were analysed ($n = 3$ biologically independent animals). A two-tailed unpaired *t*-test was used to determine the significance of the comparisons of data. $^{****}P < 0.0001$. **i**, IHC and H&E staining of lung tissue sections from the LNP-treated mice confirmed effective PTEN editing. Scale bars, 100 µm. **j**–**u**, 1A7B13 LNPs showed enhanced efficiency in delivering IL-10 mRNA to facilitate the resolution of airway inflammation following acute LPS exposure. Data are shown as mean ± s.e.m. ($n = 5$ biologically independent animals for 'Healthy mice' group, $n = 6$ biologically independent animals for 'PBS', 'DOTAP' and '1A7B13' groups, one-way ANOVA). **j**, C57BL/6 mice were administered with PBS, or LNPs encapsulating IL-10 mRNA (0.5 mg kg^{-1} IL-10 mRNA, i.v., three doses on days −1, 0 and 1). High-dose LPS (10 µg) was given on day 0 through OPA as exposure. Mice were killed on day 2 for analysis. **k**, Body weight changes of LPS-exposed mice over time. **l**, Percentage changes in body weight of LPS-exposed mice on day 2 versus baseline. **m**, Representative H&E-stained lung tissue sections. Scale bar, 100 µm. **n**, Quantitative analysis of inflammation scores derived from H&E-stained lung sections. **o**, Total number of cells collected from the lungs. **p**, IL-10 mRNA levels in the lung tissues. **q**, IL-10 protein concentrations in the BALF of the lungs. **r**, Total protein concentrations in BALF. **s**–**u**, mRNA expression levels of TNF (**s**), IL-1β (**t**) and IL-6 (**u**) in lung homogenates. Data are shown as mean ± s.e.m. One-way ANOVA followed by Dunnett's post hoc test was used for multiple comparisons. $^{*}P < 0.05$; $^{**}P < 0.01$; $^{***}P < 0.001$; $^{****}P < 0.0001$; ns, not significant, $P > 0.05$. Illustrations in **b** (right), **f** and **j** created in BioRender; Siegwart, D. https://BioRender.com/4609qdm (2025).

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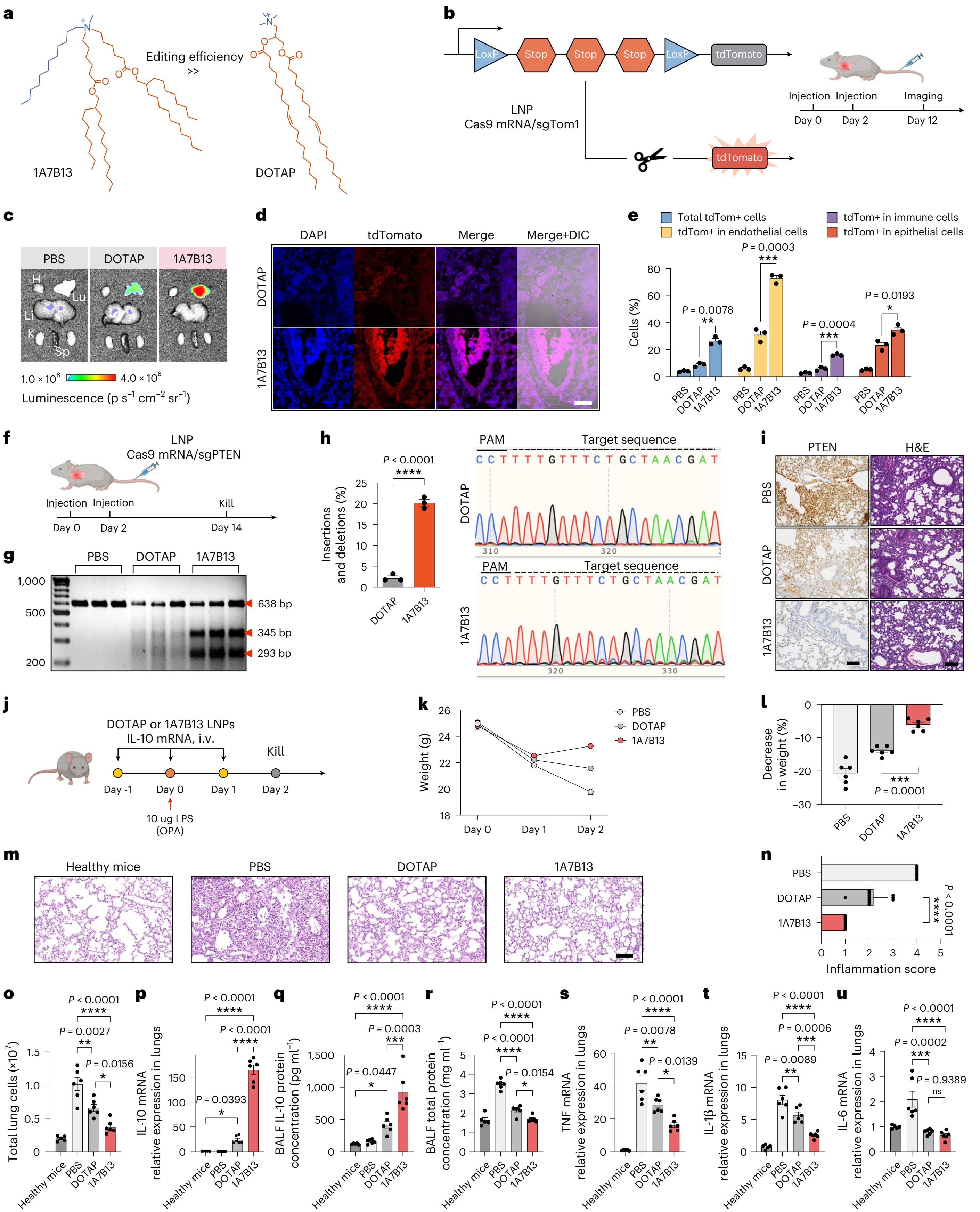

Multiple local administration strategies, including nebulization[14,16] to deliver liquid and dry powder aerosols via inhalation[15], face hurdles as they must traverse mucus and other physical barriers. Systemic administration approach remains a highly viable option for treating lung diseases with high morbidity and mortality, provided that developed carriers are sufficiently selective, efficient and tolerated. The lead 1A7B13 LNPs show exceptional transfection efficiency in both lung endothelial and epithelial cells, highlighting strong potential and advantages for treating lung diseases, including ALI associated with both the lung epithelium and endothelium, and cystic fibrosis that affects the lung epithelium. Despite these advantages, further studies are still needed to evaluate repeat dose tolerability, long-term safety and performance across diverse disease states and in larger animal models to fully define clinical potential.

Our findings deepen the understanding of the dynamic relationship between the lipid structure and lung-targeting outcomes, offering valuable guidance to researchers seeking highly effective tools for delivering mRNA and genome editors to the lungs. The high efficiency achieved at low doses also presents a promising approach to minimize the toxicity of lung-targeting LNPs. LuT lipids introduced a new path to conceptualize nanoparticle delivery by designing molecules to enhance endosomal escape and specific plasma protein interactions. Future experimental approaches to generate improved lung-targeting LNPs may include tuning linker and headgroup chemistry, as well as mapping structure-protein corona interactions across healthy and diseased lung environments. To support progression toward the clinic, key next steps will involve evaluating repeat dose safety, durability and biodistribution in large animal models, along with developing scalable manufacturing processes. Future investigation will be focused on broadening the applications of LuT lipids to treat lung genetic diseases, such as cystic fibrosis, alpha-1 antitrypsin disease or surfactant protein deficiency disorders.

Methods

Materials

4A3SC7, an ionizable lipid developed in our lab, was synthesized and purified by following published protocols[22,23]. Amines, cholesterol, Dulbecco's modified PBS, RPMI-1640 medium, fetal bovine serum and trypsin–EDTA (0.25%) were purchased from Sigma-Aldrich. DOPE, DOTAP, 1,2-dioleoyl-*sn*-glycero-3-phosphocholine (DOPC), 1,2-dioleoyl-*sn*-glycero-3-phospho-L-serine (DOPS), NBD-PE (ammonium salt) and Rho-PE (ammonium salt) were purchased from Avanti Lipids. DMG-PEG2000 was obtained from NOF America Corporation. The Quant-iT RiboGreen RNA assay kit was purchased from Life Technologies. D-Luciferin firefly, sodium salt monohydrate was purchased from Gold Biotechnology. Anti-PTEN antibody was purchased from Cell Signalling Technology (PTEN (138G6) Rabbit mAb 9559).

Synthesis and purification of the tails

The tails B9–B18 and B20–B25 were synthesized through conjugation of alkyl or alkenyl acids with hydroxyalkyl compounds. The acid was dissolved in HPLC-grade CH_2Cl_2 (80 ml, direct from the bottle) and cooled to 0 °C. Hydroxyalkyl compounds (1.0 equiv.), EDC·HCl (2.0 equiv.) and DMAP (1.2 equiv.) were added. After 30 min, the reaction was warmed to room temperature and stirred for 14 h. The solvent was removed in vacuo, and the crude product was purified by column chromatography (hexane/EtOAc = 10:1).

Synthesis and purification of LuT lipids

LuT lipids were synthesized through combinatorial reaction of amines (1A1–2A13) and alkyl or alkenyl bromides (B1–B25). B1–B8 and B19 were commercially available. B9–B18 and B20–B25 were synthesized as mentioned above. Each primary, secondary or tertiary amine was designed to consume three, two or one equivalent of alkyl or alkenyl bromides. Briefly, the amines were reacted with 3.6, 2.4 and 1.2 equivalents of alkyl or alkenyl bromides to obtain LuT lipids. All the reactions were conducted in a vial equipped with stir bar. Potassium carbonate (1.5 equiv.), amine (1.0 equiv.), and bromo-hydrocarbon (1.2 equiv.) in anhydrous DMF were added into the vial. The reaction mixture was then heated to 95 °C for 24 h before cooling to room temperature. Then the reaction mixture was concentrated in vacuum and further purified by column flash chromatography (DCM/MeOH = 10:1).

RNA synthesis

Firefly luciferase mRNA and Cre mRNA were provided by ReCode Therapeutics. Cas9 mRNA was produced using in vitro transcription as previously described[20]. Briefly, linearized plasmids encoding Cas9 protein were prepared using a PCR programme. Then, the fragment was cloned into pCS2 + MT vectors with optimized 5′(3′)-untranslated regions and poly A sequences. Cas9 mRNA was transcribed using the Invitrogen MEGAscript SP6 transcription kit (ThermoFisher) but with N^1-methylpseudouridine-5′-triphosphate instead of the typical uridine triphosphate to generate modified nucleoside-containing mRNA. Next, the mRNA was capped (Cap1) using the Capping System (Hongene).

LuT LNPs formulation and characterization

LuT LNPs were formulated as previously described[23]. In general, mRNA was dissolved in an acidic aqueous buffer (10 mM citrate buffer, pH 4). The organic phase lipid mixture containing 4A3-SC7, DOPE, cholesterol, DMG-PEG2000 and a permanently positive charged lipid was dissolved in ethanol. The organic phase and aqueous phase were mixed rapidly by vortex mixing at a volumetric ratio of 1:3, then incubated for 15 min at room temperature. The final weight ratio of total lipids and mRNA was 40:1. mRNA-loaded LNPs were dialysed against PBS using Pur-A-Lyzer midi dialysis chambers (Sigma-Aldrich) for 1 h. The size, zeta potentials and PDI were measured using dynamic light scattering (Malvern MicroV model; He–Ne laser, wavelength = 632 nm). mRNA encapsulation was quantified using the Quant-iT RiboGreen RNA assay kit (Invitrogen)[36].

Lipid fusion by FRET assay

Endosomal mimicking anionic liposomes were prepared by mixing of DOPS, DOPC, DOPE, NBD-PE and Rho-PE (molar ratio 25:25:48:1:1) in chloroform, followed by vacuum drying to give a thin lipid film. Liposomes were prepared by resuspending the thin layer in PBS (pH 7.4) and then sonicated for 20 min (10 s sonication and 10 s rest cycles) to make a final lipid concentration of 1 mM. The LuT LNPs were prepared following the procedure described above. To a black-bottom 96-well plate, 100 µl PBS (pH 5.5) was added to each well, and then 1 µl of endosomal mimicking anionic liposomes was added. LuT LNPs were added at a ratio of 10:1 (LNP/liposome). After incubating at 37 °C, fluorescence measurements (F) were conducted on a microplate reader at excitation (EX)/emission (Em) = 465/520 nm. Endosomal mimicking anionic liposomes incubated with PBS and Triton X-100 were set as negative control (F_{min}) and positive control (F_{max}), respectively. The lipid fusion (%) was calculated as $(F - F_{min})/(F_{max} - F_{min}) \times 100\%$.

Isolation of plasma proteins adsorbed on LNPs

Protein coronas were isolated following our previously reported method[31]. Briefly, LuT LNPs were added to C57BL/6 mouse plasma at a 1:1 volume ratio to make a final lipid concentration of 1 mg ml^{-1}. The mixture was incubated for 15 min at 37 °C, then loaded onto a 0.7 M sucrose cushion and centrifuged at 15,000 *g* at 4 °C for 1 h. The pellets were washed with PBS 3 times and then resuspended in 2% SDS. Excess lipids were removed using ReadyPrep 2-D Cleanup (Bio-Rad) kit. The resulting proteins were stored for further analysis.

SDS–PAGE characterization of plasma proteins

Plasma proteins isolated from the protein corona from the LNPs complex were analysed using 10% Mini-PROTEAN TGX 20 Precast Protein Gel. Then 10 µl of proteins were loaded into the gel and separated at 160 V for 1 h. The gel was visualized by staining with Bio-Safe Coomassie

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Stain for 1 h and destaining in water for another 2 h. Then, the gel was imaged by a Licor Scanner.

Preparation of plasma protein samples for mass spectrometry

For mass spectrometry, 10 µl of the plasma proteins were loaded onto a 12% Mini-PROTEAN TGX 25 Precast Protein Gel and run at 90 V for 10 min. Next, the gel was stained with Bio-Safe Coomassie Stain and then destained for 1 h. The protein bands were isolated and sliced into 1 mm^3 cubes. The cubes were submitted to the University of Texas Southwestern Proteomics Core for mass spectrometry analysis. The identified proteins were ordered based on their abundance/molecular weight (Mw). A custom Python Script was written to rank the most abundant proteins and plot them as a heat map.

Animal experiments

All animal experiments were approved by the Institution Animal Care and Use Committees of The University of Texas Southwestern Medical Center (UTSW) and were consistent with local, state and federal regulations as applicable. Mice were housed in a barrier facility with a 12 h light/dark cycle and maintained on standard chow (2916 Teklad Global). C57BL/6 mice were obtained from Charles River. B6.Cg-*Gt(ROSA)26Sor*$^{tm14(CAG-tdTomato)Hze}$/J mice (also known as Ai14 or Ai14(RCL-tdT) mice) were obtained from the Jackson Laboratory (007914) and bred to maintain homozygous expression of the Cre reporter allele.

In vivo Luc mRNA delivery

Six- to eight-week-old C57BL/6 mice, with weights of 18–20 g, were intravenously administered with Luc mRNA LNPs at a dose of 0.1 mg kg^{-1} mRNA. Then 6 h after injection, the mice were injected with D-luciferin (150 mg kg^{-1}, intraperitoneal). Mice were anaesthetized in a ventilated anaesthesia chamber with 2.5% isoflurane in oxygen and imaged by AMI imaging system (Spectral Instruments Imaging AMI-HTX). The main organs were collected and imaged after the live animal imaging. Tissue selectivity was quantified as follows:

$$\%_{\text{organ}} = \frac{\text{Bioluminescence}_{\text{organ}}}{\sum_{\text{all organs}} \text{Bioluminescence}_{\text{organ}}}$$

In vivo Cre mRNA delivery

Six- to eight-week-old tdTom mice (Ai14 mice), with weights of 18–20 g, were intravenously administered with Cre mRNA LNPs at a dose of 0.3 mg kg^{-1} mRNA. Then 48 h after injection, the mice were killed, and main organs were imaged by AMI imaging system (Spectral Instruments Imaging AMI-HTX).

In vivo Cas9 mRNA/sgRNA delivery

LNPs encapsulating Cas9 mRNA and sgPTEN were intravenously injected into C57BL/6 mice (at a dose of 0.75 mg kg^{-1} total RNA, twice a week). Then 10 days after injection, mice were killed. Lungs were isolated, and genomic DNA was collected with PureLink Genomic DNA Mini Kit (Thermo Fisher). After obtaining the PTEN PCR products, a T7E1 assay was performed to confirm gene editing efficacy using the established manufacturer protocol (NEB). Indel (%) was calculated as $100 \times (1 - (1 - \text{fraction cleaved})^{0.5})$, where the fraction cleaved is defined as (Fragment 1 + Fragment 2)/(Fragment 1 + Fragment 2 + Parent fragment). Furthermore, the collected lungs were fixed for 24 h and then embedded in paraffin. Then the specimens were sectioned and stained for H&E and IHC by Molecular Pathology at the UTSW. Briefly, the fixed lungs were embedded in paraffin and sectioned at 4 µm. For the PTEN staining, the slides were first deparaffined and then incubated with anti-PTEN antibody (PTEN (138G6) Rabbit mAb 9559, Cell Signalling). The slides were then processed following the standard protocol by Molecular Pathology Core at UTSW.

For Cas9 mRNA/sgTom1, tdTomato mice (Ai14) were injected with Cas9 mRNA/sgTom1 encapsulating LNPs. Then 12 days after injection, mice were killed, and the main organs were isolated and imaged using AMI imaging system (Spectral Instruments Imaging AMI-HTX). The lung tissues were collected and fixed at 4 °C and then were equilibrated in 18% sucrose before freezing in OCT. The cryo-section slides were washed with PBS. Afterwards, ProLong Gold Mountant was mounted with DAPI (4,6-diamidino-2-phenylindole) to visualize the nucleus. The slides were imaged using confocal microscopy (Zeiss LSM 700) and analysed by ZEN 2010 software v.6.0.62 (Carl Zeiss MicroImaging).

Flow cytometry

The flow cytometry analysis followed our published protocols[18]. Briefly, single-cell suspensions of lung tissues were prepared and analysed by flow cytometry to determine the proportion of tdTom$^+$ cells in each cell type. The lung tissues from mice were isolated and digested by 1× lung digestion media (RPMI supplemented with 2% wt/vol BSA, 300 U ml^{-1} collagenase and 100 U ml^{-1} hyaluronidase) at 37 °C for 1 h. The digested tissues were homogenized and filtered through a 70 µm cell strainer to remove the undigested organ fragment. After centrifugation and removal of the supernatant, the cell pellets were resuspended in red blood cell lysis buffer (BioLegend, 420301) for 5 min. After incubation, cell staining buffer (BioLegend) was added to stop the red blood cell lysis, and the cell suspension was then filtered and centrifuged again. The resuspended cells were incubated with antibody cocktail composed of Live/Dead Ghost Dye Red 780, Pacific Blue-conjugated anti-mouse CD45, Alexa Fluor 488-conjugated anti-mouse CD31 and Alexa Fluor 647-conjugated anti-mouse EpCAM antibodies (BioLegend) for 30 min at 4 °C in dark. Cells were then analysed by Becton Dickenson LSR Fortessa flow cytometer, and data were analysed using FlowJo software version 7.6.

In vivo IL-10 mRNA delivery

Male C57BL/6 mice weighing 18–20 g were selected for this study due to their higher susceptibility to inhaled endotoxin-induced airway and systemic inflammatory responses compared to female mice[46]. Mice received intravenous (i.v.) administration of PBS, 1A7B13 LNPs or DOTAP LNPs encapsulating IL-10 mRNA (0.5 mg kg^{-1}) on days −1, 0 and 1. On day 0, a high dose of LPS (10 µg in 50 µl PBS) derived from *Escherichia coli* (O111:B4, Sigma) was delivered via OPA to induce the airway inflammation. Body weights were monitored throughout the treatment period. Mice were killed on day 2. BALF was collected twice by gentle perfusion of the lungs with PBS. The total protein levels in the BALF supernatant were determined using a Pierce BCA Protein Assay Kit (Thermo Scientific). IL-10 protein concentrations in the BALF were measured using a Mouse IL-10 ELISA Kit (ab255729, Abcam), following the manufacturer's protocols.

Lung inflammation score analysis

Lung tissues were perfused, fixed in 4% paraformaldehyde, sectioned and stained with H&E for histological evaluation. Inflammation scores measured the inflammatory cell infiltration in the primary bronchi, alveoli and surrounding major blood vessels using the following criteria: 0, no cells; 1, minimal inflammatory cells; 2, unevenly distributed inflammatory cells; 3, abundant, evenly distributed inflammatory cells; 4, significant clustering of inflammatory cells. The average inflammation score was calculated for each experimental group. All blinded assessments were conducted independently by an expert pathologist to ensure unbiased and accurate evaluation.

Quantitative PCR

Total RNA was isolated by adding 800 µl of ice-cold TRIzol reagent (Invitrogen, 15596026) to approximately 20–30 mg of lung tissues, followed by incubation at room temperature for 5–10 min with intermittent vortexing. Chloroform (160 µl, one-fifth of the TRIzol volume) was added, and samples were vigorously mixed and incubated for 2 min at room temperature. Tissue lysates were centrifuged at 12,000 *g* for 15 min at 4 °C. The aqueous phase was carefully transferred to a new tube and mixed with an equal volume of 70% ethanol. RNA was subsequently purified using the RNeasy Mini Kit (Qiagen, 74104), following the manufacturer's protocol. RNA was eluted in 30–50 µl of nuclease-free water, and the concentration was determined using a NanoDrop 1000 Spectrophotometer (Thermo Fisher Scientific). Complementary DNA was synthesized from 1 µg of total RNA using the iScript Reverse Transcription Supermix (BioRad, 1708841) on a SimpliAmp Thermal Cycler (Thermo Scientific, A24811). Quantitative PCR was performed to assess target gene expression using a QuantStudio 5 Real-Time PCR system (Applied Biosystems, A28140) and iTaq Universal SYBR Green Supermix (BioRad, 1725121). Mouse-specific primers used for quantitative PCR are listed in Supplementary Table 5. Primer specificity and efficiency were validated by melt curve analysis. Relative gene expression was calculated using the $2^{-\Delta\Delta Ct}$ method, with actin serving as the internal reference gene.

In vivo toxicity evaluation

Female C57BL/6 mice, with weights of 18–20 g, were randomly divided into 4 groups, $n = 4$ per group. 1A7B13 LNPs were intravenously injected to mice (Luc mRNA, 0.3 mg kg^{-1}). LPS (5 mg kg^{-1}), considered as the positive control, was intraperitoneally injected. PBS was intravenously injected as the negative control. Then 24 h and 48 h after administration, whole blood from the mice was collected in Becton Dickenson Microtainer tubes, and the serum was separated after 15 min of centrifugation. The liver function (AST and ALT) and renal function (BUN and CREA) were tested by the UTSW Metabolic Phenotyping Core.

Statistical methods

Statistical analyses were performed using GraphPad Prism v.10 (GraphPad Software). Data are presented as means ± s.e.m. and statistically analysed using GraphPad Prism software version 10. One-way analysis of variance (ANOVA) followed by Dunnett's post hoc test was used for multiple comparisons. A two-tailed unpaired *t*-test was used to compare differences between two groups. $P < 0.05$ was considered to represent statistical significance.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

Source data are provided with this paper. All other data are available from the corresponding author upon reasonable request.

Code availability

The code used for the dimensionality reduction of LuT LNPs can be found on GitHub at https://github.com/willemill/Lung-targeting-LNP-analysis.

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Author contributions

Z.T., X.W., and D.J.S. designed the research. Z.T., X.W., S.C., W.M., E.G., Y.-C.S., A.P.B., S.D., X.B., A.V., X.L., S.M., Y.S., M.K., Y.X., S.W., B.M.E., and L.F. performed the experiments. J. L. assisted in verifying the statistical analyses. D.J.S. supervised the experiments and managed the project. Z.T., X.W., and D.J.S. wrote the manuscript. All the authors were involved in the data analyses and the improvement of the manuscript.

Competing interests

A provisional patent application covering compounds and compositions for targeting lung has been filed by UTSW naming D.J.S, Z.T. and X.W. as inventors. D.J.S. discloses financial interests in ReCode Therapeutics, Signify Bio, Jumble Therapeutics and Tome Biosciences. D.J.S. is a member of the scientific advisory board of ReCode Therapeutics, which has licensed intellectual property from UTSW. The remaining authors declare no competing interests.

Additional information

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Software and code

Policy information about availability of computer code

Data collection	Agilent MR400 and Bruker AN400 (see https://www.utsouthwestern.edu/labs/nmr/instruments-rates/ for details). Malvern Nano Zetasizer Aura software v.4.5.0 (Spectral Instruments Imaging) Microplate reader (TECAN Infinite® 200 PRO) ZEN x64 software version 1.1.0 (Carl Zeiss Microscopy GmbH) BD FACSDiva software version 8.0.1 (BD LSRFortessa)
Data analysis	Mestrelab's MestReNova 12.0.3-21384. GraphPad Prism 10.2.3 (GraphPd Software) Aura software v.4.0.7 (Spectral instruments imaging) TIDER analysis tool (no version information available)(analyzing indels and HDR by decomposition)(http://shinyapps.datacurators.nl/tide/) CRISPResso2 (analyzing NGS result to evaluate the editing efficiency)(http://crispresso2.pinellolab.org/submission) FLOWJO software version 7.6 (FLOWJO) ZEN 2010 software version 6.0.62 (Carl Zeiss MicroImaging GmbH) Python 3.10 (Vscode)

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Recruitment	Not applicable.
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Sample size	Sample sizes were determined using statistical power calculations.
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Randomization	For in vivo study, animals were randomly sorted into treatment and control groups.
Blinding	Data collection and analyses for some experiments were conducted by separate individuals. In some cases, these collectors/analyzers were not aware which samples corresponded to which experimental groups at the time of data collection and analysis. Scored histological sections and tissue analyses were conducted in a blinded manner.

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Materials & experimental systems

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☒	☐ Palaeontology and archaeology
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Methods

n/a	Involved in the study
☒	☐ ChIP-seq
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☒	☐ MRI-based neuroimaging

Antibodies

Antibodies used

Anti-CD45 (Pacific Blue-conjugated)(#103126), Anti-EpCam (Alexa Fluor 647-conjugated)(#118212) and Anti-CD31 (Alexa Fluor 488-conjugated anti-mouse)(#102414) were purchased from BioLegend. Anti-PTEN antibody was purchased from Cell Signaling Technology (PTEN (138G6) Rabbit mAb #9559).

Validation

The antibody was validated internally validated using positive and negative control samples.
Validation/citation data are also available on the manufacturer's websites or CiteAb's website:
Anti-CD45 (Pacific Blue-conjugated)(#103126), https://www.biolegend.com/en-us/products/pacific-blue-anti-mouse-cd45-antibody-3102
Anti-EpCam (Alexa Fluor 647-conjugated)(#118212), https://www.biolegend.com/en-us/products/alexa-fluor-647-anti-mouse-cd326-ep-cam-antibody-4973
Anti-CD31 (Alexa Fluor 488-conjugated anti-mouse)(#102414), https://www.biolegend.com/en-us/products/alexa-fluor-488-anti-mouse-cd31-antibody-3091
Anti-PTEN (PTEN (138G6) Rabbit mAb #9559), https://www.cellsignal.com/products/primary-antibodies/pten-138g6-rabbit-mab/9559

Animals and other research organisms

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Laboratory animals

Mice were housed in a barrier facility with a 12 h light/dark cycle and maintained on standard chow (2916 Teklad Global). The temperature range for the housing room is 22 ºC and the humidity range is 35%-60% (average is around 50%).C57BL/6 mice were obtained from the UTSW Mouse Breeding Core Facility. B6.Cg-Gt(ROSA)26Sortm14(CAG-tdTomato)Hze/J mice (aslo known as Ai14 or Ai14(RCL-tdT)-D mice) were obtained from The Jackson Laboratory (007914) and bred to maintain in the UTSW Animal Facility.

Wild animals

The study did not involve the use of wild animals.

Reporting on sex

Both male and female mice were used. The sex of the animals was not further considered because LNP-mediated nucleic acid delivery data did not show bias toward one gender in experiments. However, only male mice were used in the acute lung injury study.

Field-collected samples

The study did not involve samples collected from the field.

Ethics oversight

C57BL/6 and Ai14 mice experiments were approved by the Institution Animal Care and Use Committees of The University of Texas Southwestern Medical Center and were consistent with local, state, and federal guidelines as applicable.

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Methodology

Sample preparation	For Cre and Cas9-mediated gene editing in TdTOM reporter mice experiment, mouse lungs were collected; and the tissues were placed into ice cold PBS. The lung tissue was then cut into small pieces and transferred into a 50 mL tube containing 10mL of 1X lung digestion media [RPMI supplemented with 2% wt/vol BSA, 300 U/mL collagenase, 100 U/mL hyaluronidase]. The 50mL tube was then incubated at 37 °C for 1 hr while shaking at 180 rpm. After incubation, the homogenized lung cell solution was pipetted up and down several times to remove cell clumps and finally filtered through a 70-micron cell strainer into a new 50 mL falcon tube. The filter was washed with 10 mL cell staining buffer. The sample was then centrifuged at 10 rpm for 5 minutes. The supernatant was removed, and the cell pellets were resuspended in red blood cell lysis buffer at room temperature for 5 min. Next, 10mL of cell staining buffer was added to stop the red blood cell lysis. Then the cell suspension was filtered and centrifuged at 1200rpm for 5 minutes. The supernatant was removed, and the RBC free cell pellets were resuspended in certain volume of cell staining buffer. The single-cell suspensions were incubated with antibody cocktail composed of Live/Dead Ghost Dye Red 780, Pacific Blue-conjugated anti-mouse CD45, Alexa Fluor 488-conjugated anti-mouse CD31, and Alexa Fluor 647-conjugated anti-mouse EpCAM antibodies (BioLegend) for 30 min at 4°C in dark. Next, the cell pellets were washed with cell staining buffer to remove excess antibodies, and resuspended in 200-300 uL of cell staining buffer. Cells were then analyzed by Becton Dickenson (BD) LSR Fortessa flow cytometer and data were analyzed using FlowJo software version 7.6.
Instrument	BD FACSDiva software version 8.0.1 (BD LSRFortessa)
Software	FLOWJO software version 7.6 (FLOWJO)
Cell population abundance	We used toTomato (Ai14) mice to evaluate tissue specific gene editing efficiency via detecting the tdTomato mean fluorescence intensity.
Gating strategy	Gates for TdTom+ in cell types were drawn based control mice. Gating strategy were provided in SI Figures.

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